Physik II — FS 2024 — Prof. R. Wallny — Serie 4

Solution

Exercise 1: Understanding vector equations and vector operators

a) $\vec{\nabla} \times \vec{\nabla}\psi = \vec{0}$:

$$\vec{\nabla} \times \vec{\nabla}\psi = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \times \begin{pmatrix} \frac{\partial\psi}{\partial x} \\ \frac{\partial\psi}{\partial y} \\ \frac{\partial\psi}{\partial z} \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial y}\frac{\partial\psi}{\partial z} - \frac{\partial}{\partial z}\frac{\partial\psi}{\partial y} \\ \frac{\partial}{\partial z}\frac{\partial\psi}{\partial x} - \frac{\partial}{\partial x}\frac{\partial\psi}{\partial z} \\ \frac{\partial}{\partial x}\frac{\partial\psi}{\partial y} - \frac{\partial}{\partial y}\frac{\partial\psi}{\partial x} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \tag{1}$$

b) $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = \vec{0}$:

$$\begin{aligned} \vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) &= \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \cdot \begin{pmatrix} \frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \\ \frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \\ \frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \end{pmatrix} \\ &= \frac{\partial^2 A_z}{\partial x \partial y} - \frac{\partial^2 A_y}{\partial x \partial z} + \frac{\partial^2 A_x}{\partial y \partial z} - \frac{\partial^2 A_z}{\partial x \partial y} + \frac{\partial^2 A_y}{\partial x \partial z} - \frac{\partial^2 A_x}{\partial y \partial z} = 0 \end{aligned} \tag{2}$$

c) $\vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \Delta\vec{A}$:

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \vec{\nabla} \times \begin{pmatrix} \frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \\ \frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \\ \frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \times \begin{pmatrix} \frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \\ \frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \\ \frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \end{pmatrix} \tag{3}$$

$$= \begin{pmatrix} \frac{\partial}{\partial y}(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}) - \frac{\partial}{\partial z}(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}) \\ \frac{\partial}{\partial z}(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}) - \frac{\partial}{\partial x}(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}) \\ \frac{\partial}{\partial x}(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}) - \frac{\partial}{\partial y}(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}) \end{pmatrix} = \begin{pmatrix} \frac{\partial^2 A_x}{\partial x^2} + \frac{\partial^2 A_y}{\partial x \partial y} + \frac{\partial^2 A_z}{\partial x \partial z} \\ \frac{\partial^2 A_x}{\partial x \partial y} + \frac{\partial^2 A_y}{\partial y^2} + \frac{\partial^2 A_z}{\partial y \partial z} \\ \frac{\partial^2 A_x}{\partial x \partial z} + \frac{\partial^2 A_y}{\partial z \partial y} + \frac{\partial^2 A_z}{\partial z^2} \end{pmatrix} - \begin{pmatrix} \partial_x^2 A_x + \partial_y^2 A_x + \partial_z^2 A_x \\ \partial_x^2 A_y + \partial_y^2 A_y + \partial_z^2 A_y \\ \partial_x^2 A_z + \partial_y^2 A_z + \partial_z^2 A_z \end{pmatrix} \tag{4}$$

$$= \begin{pmatrix} \frac{\partial^2 A_x}{\partial x^2} + \frac{\partial^2 A_y}{\partial x \partial y} + \frac{\partial^2 A_z}{\partial x \partial z} \\ \frac{\partial^2 A_x}{\partial x \partial y} + \frac{\partial^2 A_y}{\partial y^2} + \frac{\partial^2 A_z}{\partial y \partial z} \\ \frac{\partial^2 A_x}{\partial x \partial z} + \frac{\partial^2 A_y}{\partial z \partial y} + \frac{\partial^2 A_z}{\partial z^2} \end{pmatrix} - \begin{pmatrix} \Delta A_x \\ \Delta A_y \\ \Delta A_z \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} (\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}) - \Delta\vec{A} = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \Delta\vec{A} \tag{5}$$

d) Using the partial derivatives with respect to x, y, and z, one obtains

$$\vec{\nabla} f = \begin{pmatrix} -\sin(x)\ln(y)e^z \\ \cos(x)\frac{1}{y}e^z \\ \cos(x)\ln(y)e^z \end{pmatrix} \tag{6}$$

e) The divergence is

$$\vec{\nabla} \cdot \vec{a} = \frac{\partial}{\partial x}(y^2 z^3) + \frac{\partial}{\partial y}(2xyz^3) + \frac{\partial}{\partial z}(3xy^2 z^2) = 0 + 2xz^3 + 6xy^2 z \tag{7}$$

The curl is

$$\vec{\nabla} \times \vec{a} = \begin{pmatrix} \frac{\partial}{\partial y}(3xy^2 z^2) - \frac{\partial}{\partial z}(2xyz^3) \\ \frac{\partial}{\partial z}(y^2 z^3) - \frac{\partial}{\partial x}(3xy^2 z^2) \\ \frac{\partial}{\partial x}(2xyz^3) - \frac{\partial}{\partial y}(y^2 z^3) \end{pmatrix} = \begin{pmatrix} 6xyz^2 - 6xyz^2 \\ 3y^2 z^2 - 3y^2 z^2 \\ 2yz^3 - 2yz^3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \tag{8}$$

Exercise 2: Electric dipole

a) The potential of a dipole declines faster than that of a point charge, as evident from comparing $\Phi_{dipole} \propto \frac{1}{r^2}$ to $\Phi_{monopole} \propto \frac{1}{r}$. This can be explained as with increasing distance of the observer, the potentials of the two charges of the dipole q und $-q$ compensate each other. Within the xy plane (i.e. $z = 0$), $\theta = 90°$ and thus the potential equals to zero.

b) The electric field of a dipole can be calculated via $\vec{E} = -\nabla\Phi$. In spherical coordinates ∇ transforms to

$$\nabla = \left(\frac{\partial}{\partial R}, \frac{1}{R}\frac{\partial}{\partial\theta}, \frac{1}{R\sin(\theta)}\frac{\partial}{\partial\phi}\right) \tag{9}$$

which leads to

$$E_r = \frac{2p \cdot \cos(\theta)}{4\pi\epsilon_0 R^3}, \qquad E_\theta = \frac{p \cdot \sin(\theta)}{4\pi\epsilon_0 R^3}, \qquad E_\phi = 0. \tag{10}$$

The field is cylindrically symmetrical around the dipole axis as it is independent of the azimuthal angle ϕ.

Exercise 3: E-field and electric potential of a charge distribution

a) For symmetry reasons $\vec{E}(\vec{r}) = E(r)\vec{e}_r$. To calculate E at a distance r from the centre of the sphere, we consider the flux of $\vec{E}$ through a spherical Gauss surface A with Radius r around the centre of the sphere.

$$\oint_A \vec{E}(\vec{r'})\mathrm{d}A' = E(r)4\pi r^2 = \frac{Q_{in}}{\epsilon_0} \tag{11}$$

Case 1: $r \geqslant R$

$$Q_{in} = V_{in} \cdot \rho = \frac{4}{3}\pi R^3 \rho \to E_a(r) = \frac{\rho R^3}{3r^2\epsilon_0} \tag{12}$$

Case 2: $r \leqslant R$

$$Q_{in} = V_{in} \cdot \rho = \frac{4}{3}\pi r^3 \rho \to E_i(r) = \frac{\rho r}{3\epsilon_0} \tag{13}$$

b) Case 1: The E-Field outside of the sphere is decreasing quadratically with the distance as it would for a point charge.
Case 2: The E-Field inside of the sphere increases linearly because the number of charge carriers increases with r^3.

c) Case 1 (Outside of the sphere):

$$\Phi_\mathrm{a}(\vec{r}) = -\int_\infty^{\vec{r}} \vec{E}_\mathrm{a}(\vec{r'})\mathrm{d}\vec{r'} = -\int_\infty^{r} \frac{\rho R^3}{3r'^2\epsilon_0}\mathrm{d}r' = \frac{\rho R^3}{3r\epsilon_0} \qquad \text{mit} \quad \Phi(\infty) = 0 \tag{14}$$

This corresponds to the potential of a point charge.

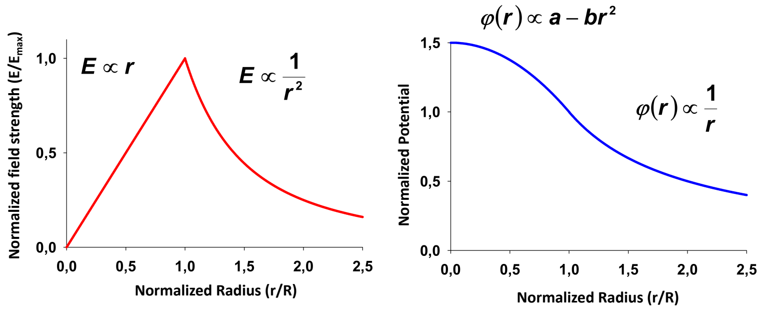
Case 2 (Inside of the sphere):
With vanishing potential at infinity one obtains the following inside of the sphere:

$$\Phi_\mathrm{i}(\vec{r}) = -\int_\infty^{\vec{r}} \vec{E}(\vec{r'})\mathrm{d}\vec{r'} = -\int_\infty^{R} \vec{E_\mathrm{a}}(\vec{r'})\mathrm{d}\vec{r'} - \int_R^{r} \vec{E_\mathrm{i}}(\vec{r'})\mathrm{d}\vec{r'} = \frac{\rho R^2}{3\epsilon_0} - \frac{\rho}{6\epsilon_0}(r^2 - R^2) \tag{15}$$

$$= \frac{\rho R^2}{2\epsilon_0}\left(1 - \frac{r^2}{3R^2}\right) \tag{16}$$

The continuity condition: $\Phi_{\mathrm{a}}(R) = \Phi_{\mathrm{i}}(R) = \dfrac{\rho R^2}{3\epsilon_0}$ is fulfilled.

d)



Exercise 4: Charged circular ring

a)

$$\Phi(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\int \frac{\rho(\vec{r}')}{|\vec{r}-\vec{r}'|}d^3r' \tag{17}$$

For an infinitely thin disk we calculate in cylindrical coordinates which simplifies the integral and gives

$$\Phi_S(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\int \frac{\sigma(\vec{r}')}{|\vec{r}-\vec{r}'|}d^2r' \tag{18}$$

with surface charge density

$$\sigma = \frac{Q}{\pi(R^2 - a^2)}. \tag{19}$$

The surface element becomes

$$d^2r' = r' \cdot d\phi' \cdot dr' \tag{20}$$

with distance (for points on the z-axis)

$$|\vec{r}-\vec{r}'| = \sqrt{z^2 + r'^2} \tag{21}$$

This results in the integral

$$\Phi_S(z) = \frac{\sigma}{4\pi\varepsilon_0}\int_a^R \int_0^{2\pi} \frac{r'}{\sqrt{z^2 + r'^2}}d\phi dr' \tag{22}$$

Due to the symmetry in ϕ, it is simplified to

$$\Phi_S(z) = \frac{\sigma}{2\varepsilon_0} \int_a^R \frac{r'}{\sqrt{z^2 + r'^2}} dr' \tag{23}$$

$$= \frac{\sigma}{2\varepsilon_0} \left[\sqrt{z^2 + r'^2} \right]_a^R \tag{24}$$

$$= \frac{\sigma}{2\varepsilon_0} \left[\sqrt{z^2 + R^2} - \sqrt{z^2 + a^2} \right] \tag{25}$$

$$= \frac{Q}{2\pi\varepsilon_0 \cdot (R^2 - a^2)} \left[\sqrt{z^2 + R^2} - \sqrt{z^2 + a^2} \right] \tag{26}$$

$$\tag{27}$$

b) The electric field $\vec{E}$ is determined by calculating the gradient of the potential: $\vec{E} = -\text{grad}\Phi$. Due to the symmetry of the arrangement $\vec{E}$ is dependent on z only:

$$\vec{E}(z) = -\frac{\sigma}{2\varepsilon_0} \cdot \frac{d}{dz} \left[\sqrt{z^2 + R^2} - \sqrt{z^2 + a^2} \right] \vec{e}_z \tag{28}$$

$$= -\frac{\sigma}{2\varepsilon_0} \cdot z \cdot \left(\frac{1}{\sqrt{z^2 + R^2}} - \frac{1}{\sqrt{z^2 + a^2}} \right) \vec{e}_z \tag{29}$$

$$= \frac{Q}{2\pi\varepsilon_0 \cdot (R^2 - a^2)} \cdot z \cdot \left(\frac{1}{\sqrt{z^2 + a^2}} - \frac{1}{\sqrt{z^2 + R^2}} \right) \vec{e}_z \tag{30}$$

c) The electric field of the point charges can be calculated using Gauss's law.

$$\oint \vec{E}_P d\vec{A} = \frac{q}{\varepsilon_0} \tag{31}$$

We make use of the symmetry of the arrangement and integrate over a spherical surface around the point charge $q = \frac{-Q}{2}$.

$$\int_0^{2\pi} \int_0^{\pi} E_P r^2 \sin\theta d\theta d\phi = \frac{-Q}{2\varepsilon_0} \tag{32}$$

$$E_P \left[2\pi r^2 (-\cos(\theta)) \right]_0^{\pi} = \frac{-Q}{2\varepsilon_0} \tag{33}$$

$$E_P \cdot 4\pi r^2 = \frac{-Q}{2\varepsilon_0} \tag{34}$$

$$E_P = \frac{-Q}{8\pi r^2 \varepsilon_0} \tag{35}$$

With the positions of the charges $(x, y = 0, z = \pm b)$ one obtains

$$\vec{E}_P(z) = \frac{-Q}{8\pi(z \mp b)^2 \varepsilon_0} (\mp \vec{e}_z) \tag{36}$$

d) The potential is obtained by integration of the electric field $\vec{E}$

$$\int_r^{\infty} \vec{E} d\vec{s} = -\left[\Phi_{P(+\infty)} - \Phi_{P(r)}\right] \tag{37}$$

$$\Phi_{P(r)} = \int_r^{\infty} \vec{E} d\vec{s} \tag{38}$$

$$= \int_r^{\infty} -\frac{Q}{8\pi r'^2 \varepsilon_0} dr' \tag{39}$$

$$= \left[\frac{Q}{8\pi r' \varepsilon_0}\right]_r^{\infty} \tag{40}$$

$$\Phi_{P(r)} = -\frac{Q}{8\pi\varepsilon_0 r} \tag{41}$$

At the positions of the point charges this results in

$$\Phi_{P(x,y,z)} = -\frac{Q}{8\pi\varepsilon_0 \sqrt{x^2 + y^2 + (z \pm b)^2}.} \tag{42}$$

where we made use of $r = \sqrt{x^2 + y^2 + z^2}$.

e) At $x = 0, y = 0, z = 0$ and $R = 2a$ the following holds for each potential:

$$\Phi_S(0) = \frac{Q}{2\pi\varepsilon_0(3a^2)} \left(\sqrt{4a^2} - \sqrt{a^2}\right) \tag{43}$$

$$= \frac{Q}{6\pi\varepsilon_0 a} \tag{44}$$

$$\Phi_P(0,0,0) = -\frac{Q}{8\pi\varepsilon_0 \sqrt{(\pm b)^2}} \tag{45}$$

$$= -\frac{Q}{8\pi\varepsilon_0 b} \tag{46}$$

For the total potential the following holds:

$$\Phi_{\text{total}} = \Phi_S(0) + 2 \cdot \Phi_P(0,0,0) \stackrel{!}{=} 0 \tag{47}$$

$$\frac{Q}{6\pi\varepsilon_0 a} - \frac{2Q}{8\pi\varepsilon_0 b} = 0 \tag{48}$$

$$\frac{1}{6a} = \frac{2}{8b} \tag{49}$$

$$a = \frac{2}{3} b \tag{50}$$

Exercise 5: Hydrogen atom

a) Because the atom is neutral, the integral of the electron's charge distribution over the whole space must equal $-e$. Evaluation of this integral will lead to an expression for ρ_0:

$$-e = \int_{\mathbb{R}^3} \rho \, dV = \int_0^{\infty} \rho(r) 4\pi r^2 dr. \tag{51}$$

Here we used the space integral in spherical coordinates and executed the angular integrals. Substitute for $\rho(r)$:

$$-e = -4\pi\rho_0 \int_0^\infty r^2 \exp(-2r/a)dr. \tag{52}$$

Using integration by parts yields

$$\int_0^\infty r^2 \exp(-2r/a)dr = \frac{a^3}{4} \tag{53}$$

and thus

$$\rho_0 = \frac{e}{\pi a^3}. \tag{54}$$

b) Because of the spherical symmetry, the electric field is in radial direction and it is the sum of the fields generated by the proton and by the electron charge cloud:

$$E = E_p + E_{\text{cloud}} = \frac{ke}{r^2} + E_{\text{cloud}}. \tag{55}$$

The field of the electron cloud at radius r is given by Gauss's law

$$\int_{\partial V} \vec{E} \cdot d\vec{A} = \frac{Q(r)}{\varepsilon_0} \tag{56}$$

where $Q(r)$ is the total negative charge within the radius r. The integral goes over the closed surface of the sphere with radius r, this surface is $4\pi r^2$. Due to symmetry the electric field is radial and, therefore, parallel to the surface element $d\vec{A}$. Hence, it follows $\vec{E} \cdot d\vec{A} = EdA$. Since the field strength only depends on the distance to the atomic nucleus, the integral in this situation can be simplified as follows:

$$\int_{\partial V} \vec{E} \cdot d\vec{A} = \int_{\partial V} EdA = E\int_{\partial V} dA = E4\pi r^2 = \frac{Q(r)}{\varepsilon_0} \tag{57}$$

Therefore, the electric field can be written as

$$E_{\text{cloud}}(r) = \frac{kQ(r)}{r^2} = \frac{Q(r)}{4\pi\varepsilon_0 r^2} \tag{58}$$

where $Q(r)$ is the negative charge enclosed within a distance r from the proton. As in a), $Q(r)$ is given by:

$$Q(r) = -4\pi\rho_0 \int_0^r r'^2 \exp(-2r'/a)dr'. \tag{59}$$

Integration by parts yields

$$\int_0^r r'^2 \exp(-2r'/a)dr' = \frac{a^3}{4} - \frac{a^3\exp(-2r/a)}{4} - \frac{a^2 r\exp(-2r/a)}{2} - \frac{ar^2\exp(-2r/a)}{2}. \tag{60}$$

Substituting in the expression for the field and taking into account that $\rho_0 = \frac{e}{\pi a^3}$, we get the final result:

$$\begin{aligned} E(r) &= \frac{ke}{r^2} + \frac{kQ(r)}{r^2} && (61)\\ &= \frac{ke}{r^2} - \frac{4ke}{a^3r^2}\left[\frac{a^3}{4} - \frac{a^3\exp(-2r/a)}{4} - \frac{a^2r\exp(-2r/a)}{2} - \frac{ar^2\exp(-2r/a)}{2}\right] && (62)\\ &= \frac{ke}{r^2}\exp(-2r/a)\left[1 + \frac{2r}{a} + \frac{2r^2}{a^2}\right]. && (63)\end{aligned}$$